STU22006 Management Science Methods – Assignment 2

Report on Optimizing Pedestrian Flow at Pearse Street

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1. Project Goals and Objectives

The primary goal of this project is to analyze and optimize the pedestrian crossing system at Pearse Street Station, a critical transit node for Trinity College Dublin (TCD). Our objective was to change the current fixed-timing signal system and propose a more flexible and efficient system that aligns with pedestrian behavior patterns.

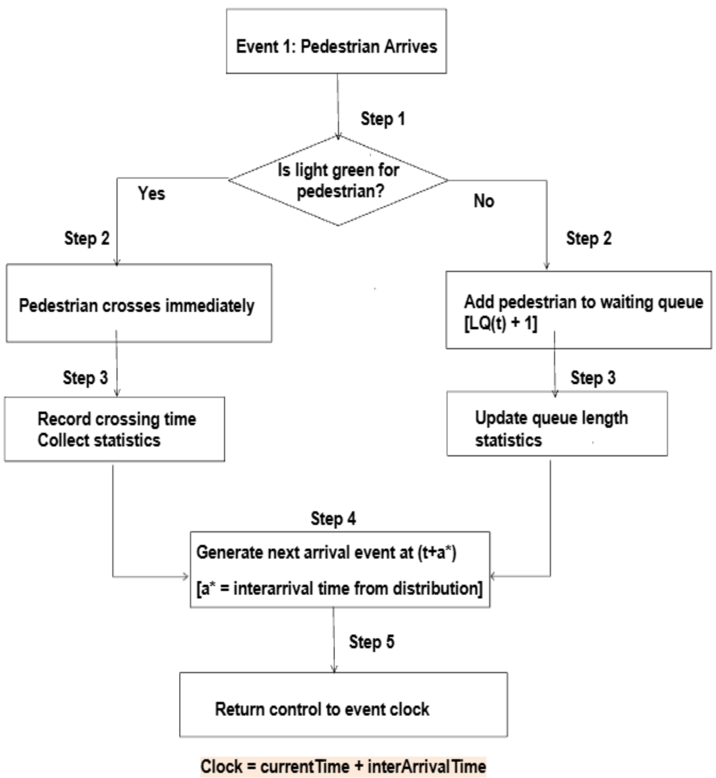
We aim to:

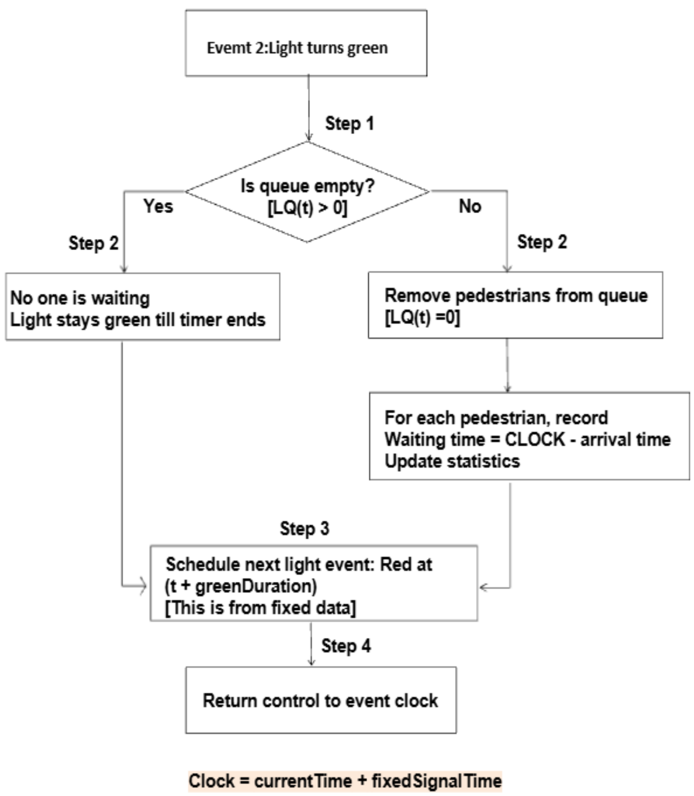
- Minimize the average pedestrian waiting time during peak periods (Lecture start and end time).
- Reduce safety risks associated with *'jaywalking'* caused by excessive delays.

2. Justification of Chosen Methods and data selection

To address the congestion at the Pearse Street crossing, this study proposes to develop a Discrete Event Simulation (DES) model embedded within a simulation-based optimization framework. Accurately modelling the unpredictable nature of pedestrians at the Pearse Street crossing (i.e., sudden arrival of large groups due to lecture finish) is difficult to capture. Therefore, we have chosen DES as the best approach to accurately model the system over time, as it can capture the transient nature of crowd during peak hours. The crossing here is seen as a queuing system: pedestrians arrive according to a non-homogeneous Poisson process with time-varying arrival rates, where inter-arrival times follow an exponential distribution with a rate that varies by time of day, and the green signal acts as a service mechanism.

2.1 Event Logic Diagram





2.2 Data Collection

2.2.1. Primary Field Observations

- Record current fixed signal timings (green/red phases, cycle length)
- During peak periods (lecture start/end times), measure pedestrian arrival rates and crossing counts per cycle.
- All data collected are based on the idealized model that pedestrians only crossed during green light; thus all recorded arrivals reflect legal crossing behavior.
- During peak periods (lecture start/end times), measure pedestrian arrival rates and crossing counts per cycle.

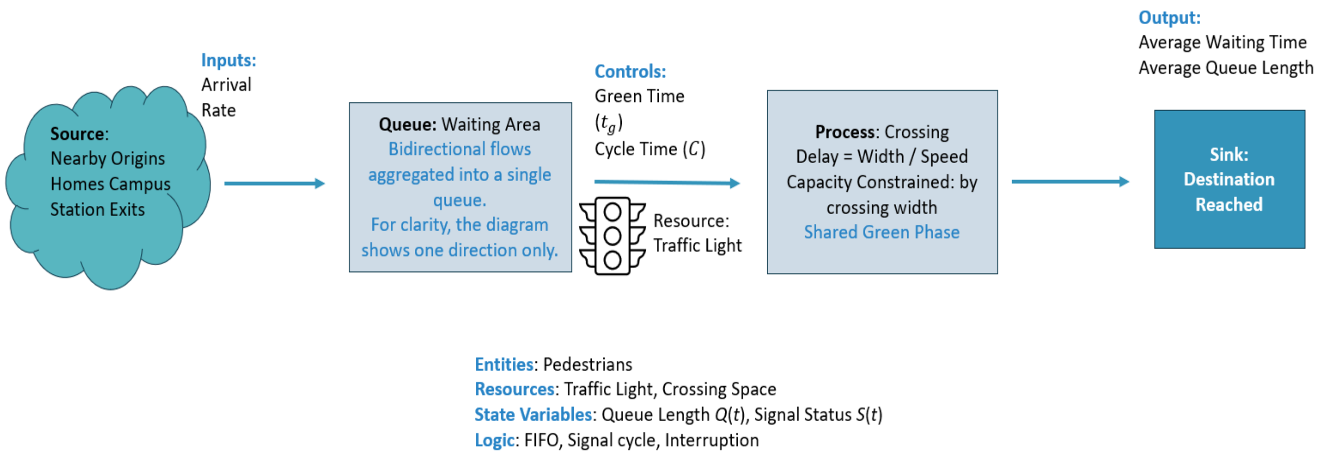
Cycle length (seconds)	99
Current green light duration (seconds)	17
Current red-light duration (seconds)	82
Average arrival rate (at Trinity main campus)	16.06060606
Average arrival rate (at Goldsmith)	18.295
System capacity (maximum number of people in waiting area)	40
Average queue length (at Trinity main campus)	26.5
Average queue length (at Goldsmith)	30.188

2.2.2. Secondary research

The UK charity living streets recommends that the pedestrian waiting time at signalized crossings should be 30 seconds at most, as research shows longer delays tempt pedestrians to cross against the signal (Living streets, 2012).

3. Modeling and Simulation Accomplishments

3.1 Conceptual Model



3.2 Specification Model

3.2.1 System Definition

Type: Signalised pedestrian crossing modelled as a single aggregated queue

Flow assumption: Bi-directional pedestrian flows (East→West and West→East) share the same green phase. Their arrivals are combined into one equivalent arrival process λ(t).

Objective: The objective of this model is to minimize pedestrian waiting time. To quantitively achieve the goal, we adjust the green duration.

3.2.2 Model Parameters and Decision variable

Decision variable works as the control variable, which could be adjusted to improve performance. Also, it influence state variables and then determine waiting time.

Arrival Rate $\boldsymbol{\lambda(t)}$

Estimated from our field data (n = 11 cycles).

Service Rate $\boldsymbol{\mu}$

Service occurs only during green light. Service rate μ (pedestrians/second) estimated from the same filed observations. During red light, μ = 0

Cycle Time $\mathbf{C}$

Fixed cycle length: C = red duration+ green duration

Current setting: C = 82+ 17 =99s

Green Duration $\mathbf{t_g}$ **(Decision variable)**

Red duration is then C − t_g

3.2.3 State Variables

State variables describe the current condition of the system and change over time, they are affected by arrivals and light duration adjustment.

Q(t): number of pedestrians waiting to start crossing at time t

S(t): signal state at time t, S(t) ∈ {Red, Green}

3.2.4 State Dynamics

Let$\Delta \mathrm{t} = 1\mathrm{s}$. *Let A(t) be the number of arrivals during* $(\mathrm{t}, \mathrm{t}+\Delta\mathrm{t})$, *with*$E[A(t)] = \lambda(t)\Delta t$

If signal = red: $\mathrm{Q}(\mathrm{t}+\Delta\mathrm{t}) = \mathrm{Q}(\mathrm{t}) + \mathrm{A}(\mathrm{t})$ *(Queue grows)*

If signal = green: $\mathrm{Q}(\mathrm{t}+\Delta\mathrm{t}) = \max(0, \mathrm{Q}(\mathrm{t}) + \mathrm{A}(\mathrm{t}) - \mu\ \Delta\mathrm{t})$ *(Queue clears)*

These equations illustrate the queue-update logic conceptually. The simulation is implemented as an event-driven DES using a Future Event List, not a fixed time-step loop.

3.3 Computational Model

We developed a discrete-event simulation (DES) of the Pearse Street crossing to model pedestrian queuing under the current signal timing and test alternative configurations. Parameters were calibrated using field-collected data. To account for randomness, we ran 100 replications per scenario and used a t-test to confirm that the improvements from the optimized signal timing is statistically significant. The simulation was implemented using R**,** enabling us to track individual pedestrian agents and their interactions with the signal.

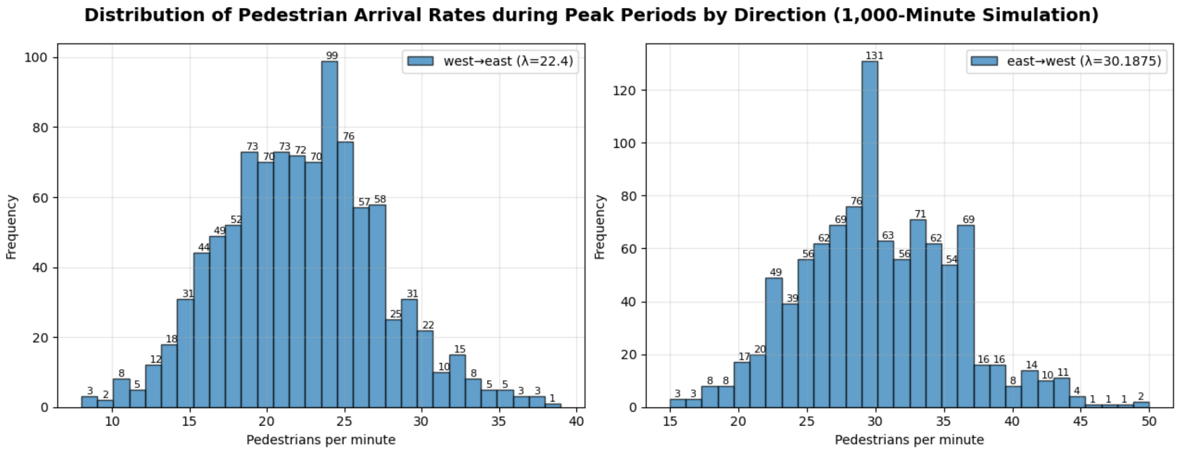


Figure 1. Histogram of Pedestrian Arrival Rate by Direction

The histogram shows a slightly right-skewed distribution. For west-to-east flow there are 22.4 arrivals per minute on average, while for east-to-west flow there are 30.2 arrivals per minute on average. There is higher number of people in queue on average on the Goldsmith side. We assume the maximum capacity of system is 40 pedestrians in waiting area. However, there are multiple datapoints showing arrivals exceeding 40 pedestrians per minute, indicating periods where queues accumulate rapidly.

4. Discussion of Results

The following table summarizes the results for average waiting time, utilization, and queue length under the current and optimized configurations. Utilization (ρ) is defined as the ratio of arrival rate to effective service rate, indicating how close the system is to its maximum capacity.

	Current	Optimised	
Average Waiting time (seconds)	69.7	29.8	↓ 57.2%
Utilization (ρ)	1.000	0.654	
Average Queue length	26.7	12.35	↓ 53.7%

Under the current configuration, pedestrians wait an average of 69.7 seconds, with a queue length of 26.7 and a utilization factor of ρ = 1. This means the arrival rate matches the effective service rate, so the queue stays the same size. After optimization, average waiting time drops to 29.8 seconds, which meets the Living Streets threshold for acceptable pedestrian delay. Queue length decreases to 12.35, reflecting a 53.7% reduction in congestion. The utilization factor falls to ρ = 0.654, meaning the system now has spare capacity to handle demand surges such as large lecture dismissals.

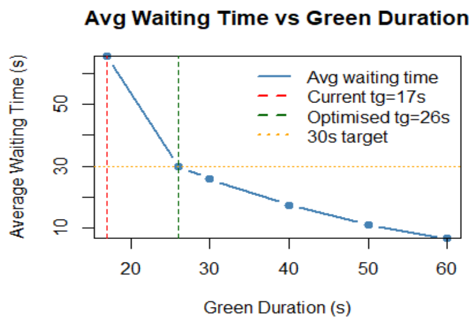


Figure 2: Waiting Time vs. Green

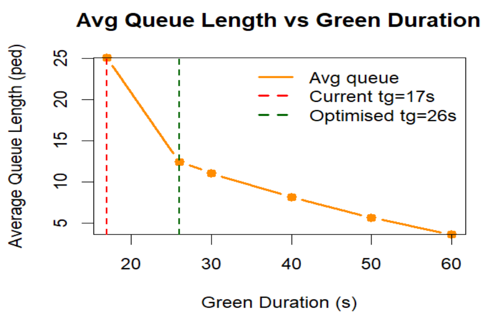


Figure 3: Queue Length vs. Green

Figure 2 shows the diminishing relationship between green light duration and average waiting time. Increasing the green light duration from 17 seconds to 26 seconds cuts the waiting time by approximately 57%. However, increasing the time beyond 30 seconds yields progressively smaller gains. The 26 second optimal sits at the inflection point of the curve, meeting the 30 second target.

Figure 3 follows the same pattern as Figure 2. At 17 seconds the queue has more than 25 people in it (this is close to the 40-person limit), leaving no spare capacity in case of a spike in demand. At 26 seconds however, this falls to around 13 people, thus restoring buffer capacity

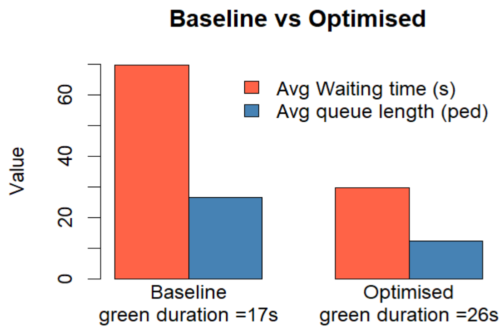


Figure 4 summarizes the improvement across all three key metrics: waiting time, queue length, and utilization. Under the optimized timing, both waiting time and queue length have roughly been halved.

Figure 4: Comparison of Key Performance Metrics

Beyond waiting times and queue lengths, another key finding was the prevalence of pedestrians crossing during the red light. This occurs primarily when red light lasts for more than 1 minute. This finding aligns with the Living streets research cited in section 2.2, which identifies 30 seconds as the threshold beyond which pedestrian compliance declines. Under the optimized configuration, the red phase is 73 seconds (with a 26-second green), which reduces but does not eliminate this risk. Therefore, we recommend using an adaptive system instead of fixed durations of signals, where the signal adjusts dynamically according to pedestrians and vehicles in queue.

5. Iterative Learning and Model Refinements

Several assumptions and approaches were revisited during the development of the model, leading to key refinements.

Firstly, train arrivals at Pearse Street DART station were initially hypothesized as primary drivers of pedestrian demand. However, field observations revealed that DART arrivals have a weak effect on arrival rates. Unlike students, who tend to move in concentrated groups toward a common destination, DART passengers disperse in multiple directions upon exiting the station. Consequently, train data was retained only as a secondary contextual factor rather than a core input to the pedestrian arrival model.

Secondly, a limitation of this study is that the optimized signal timing was evaluated purely from a pedestrian perspective. Approximately half of pedestrian red phases occurred when no vehicles were present, largely due to low left-turn traffic volumes. This suggests that reallocating this unused green time could improve both pedestrian and vehicular flows simultaneously. Future work could explore this trade-off by modelling two-way optimization and investigating extended green durations to manage lecture-end demand surges.

Thirdly, we considered a pedestrian-activated system that would reduce the green light countdown upon button press. However, pedestrian buttons in busy urban areas often have a limited or "placebo" effect to prevent traffic disruption (TheJournal.ie, 2014). Introducing such an override at Pearse Street would likely disrupt Dublin bus schedules during peak hours, making this approach infeasible.

6. Recommendations for the Future

Based on our findings, our recommendations for Dublin City Council are as follows:

i. **Dynamic pedestrian signals:** The traffic light system should incorporate a priority period aligned with lecture end times to provide students sufficient time to cross safely. We recommend increasing the green light duration to 26 seconds per cycle during peak windows, as our simulation results show this duration effectively clears the queue.

ii. **Sensor Deployment:** To adapt to the schedule in real time, we recommend installing overhead infrared or LiDAR sensors to continuously measure sidewalk occupancy. These sensors would allow the signal controller to trigger green phase extensions dynamically when occupancy exceeds a set threshold, making the system more responsive on irregular days such as exam periods or public events. This approach has been successfully piloted by Transport for London (TfL) at 18 crossings, where green is displayed as the default setting for pedestrians and changes to red only when sensors detect an approaching vehicle. After a nine-month trial, pedestrians saved 1.3 hours per day at the average crossing and showed a 13% increase in compliance, with virtually no impact on traffic. These findings demonstrate the effectiveness of sensor driven adaptive control in responding to real time pedestrian demand (TfL, 2021).

iii. **Rail API integration:** Although train arrivals were found to be a weak predictor of pedestrian demand overall, integrating Irish Rail's live departure data could still provide a useful safety buffer on days with high variability. For instance, when trains are delayed, they tend to release an unusually large number of passengers simultaneously, which could temporarily exceed normal pedestrian flow patterns. Access to real-time rail data would allow the signal system to anticipate and respond to such surge events.

References

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